নকশা Naksha

G2G Land Data Interoperability Engine

Secure, controlled data sharing among government land agencies

One parcel. One identity. Every land agency looking at the same data.

Naksha is a Government-to-Government (G2G) land-data interoperability engine. It lets AC Land, Sub-Registry, Union Land, DLRS and DC offices share parcel information through a common parcel identity and standard data models under role-based access, while **each agency remains the owner of its official records**. The Chrome side panel is the first interface; API access for other government systems is planned.

1 Problem statement

THE PROBLEM

Land data sits in separate systems of **five government agencies**. Each owns its records, but each uses its own identifier, format, units, language and survey era, and there is **no secure, controlled channel** to share or reconcile them. When an officer acts, other agencies' records are invisible, so officers cross-check paper by hand and records drift apart.

For example: a deed registered with no mutation opened · LD Tax paid by a non-recorded holder · deed area beyond the surveyed polygon · an acquisition notice invisible to the offices that update records.

Goal. Secure G2G data sharing: one parcel identity, standard data models, role-based access, agencies keep ownership.

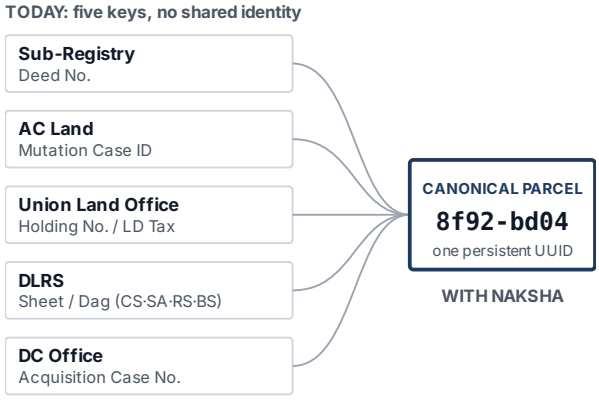
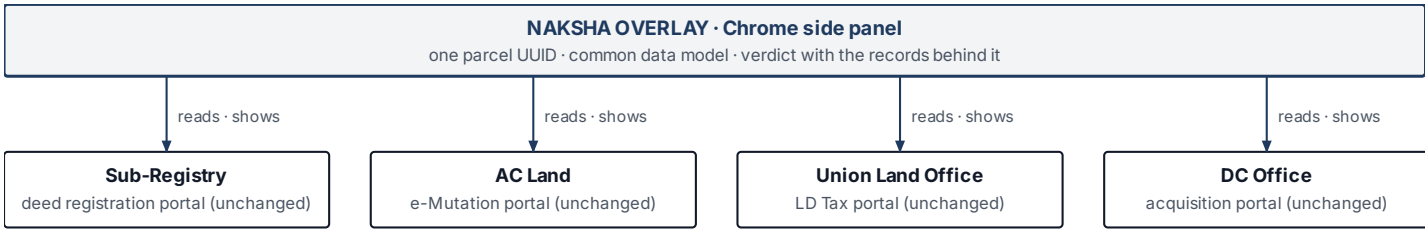


Fig 1. Five agencies, five unrelated keys, one canonical identity.

2 Solution: a G2G interoperability engine; the browser extension is one interface

Naksha links agency-specific records through a **common parcel UUID**, normalises them with **standardised data models**, and limits each officer to what their role permits through **secure, role-based access**. Agencies keep their official records; Naksha reads through adapters and keeps only the identity, cross-references and an audit trail.



Each agency stays the source of record. Naksha reads, links and compares; it does not replace or modify agency systems.

Fig 2. The Chrome side panel is one interface to the engine; agency portals stay unchanged.

NAKSHA IS

- a G2G interoperability engine
- one parcel identity and data model
- role-based, audited access

NAKSHA IS NOT

- a new registry or replacement system
- a decision-maker (no ownership rulings)
- a fraud detector, AI model or public service

3 Unique selling points

1. **One identity, one data model.** A common parcel UUID and standard models link five agencies and four survey eras.
2. **Normalisation built for Bangladesh.** Bangla/English digits, শতক · বিঘা · কাঠা · একর, phonetic name matching, CS→BS lineage.
3. **Adapter-based integration.** Each agency connects through an adapter matching its capability (API, read replica, file export, officer's open page); no change to ministry software. Prototype: synthetic data; live integrations planned.
4. **Secure, role-based access.** No Naksha login; identity is read from the portal session (simulated now; SSO in production).
5. **Explainable and auditable.** Deterministic rules, no black-box model; every action sealed in a signed ledger.

4 Who are our users, and what do they use it for?

Government agency	Use case	What Naksha provides
Sub-Registry Office	Land transfer	Checks a deed against mutation, area and survey records of other agencies.
Upazila Land Office (AC Land)	e-Mutation; land disputes	Verified deed, CS→BS lineage, name-match score; record history for disputes.
Union Land Office	Land development tax	Shows the recorded holder, land class and neighbouring plots.
DLRS	Survey and record correction	Compares deed claims with surveyed polygons and Khas land.
DC Office	Land acquisition	Shows the recorded holder, surveyed polygon and acquisition notices.

All users are authorised government officers. Naksha flags discrepancies; the responsible officer decides.

5 Workflow: one click from the officer portal to a verified, shared view

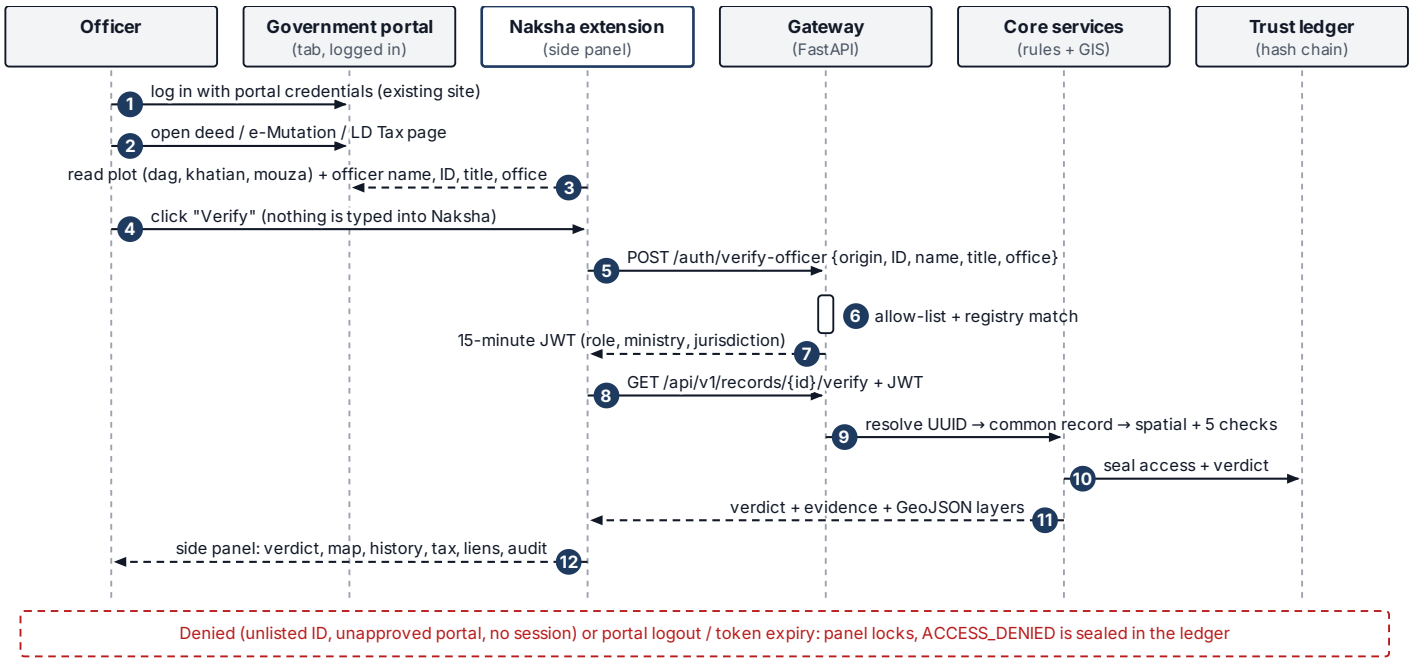
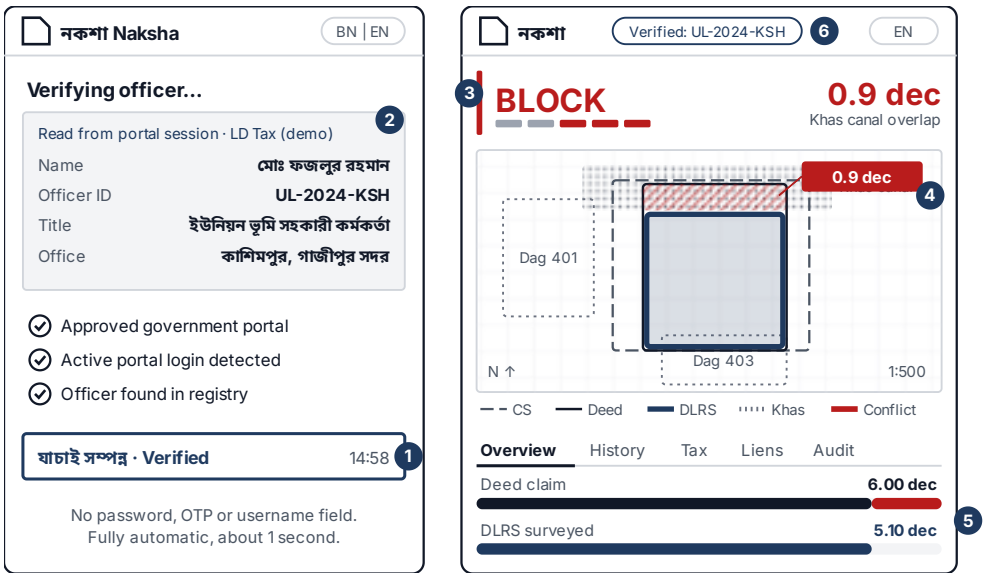


Fig 3. Sequence diagram. Solid = request, dashed = response or read. The officer types nothing into Naksha.

6 Officer experience in the browser extension



1. **One click, no new login**
   "Verify" appears only after the portal session is found.
2. **Identity is read, not typed**
   Name, ID, title, office come from the open portal page.
3. **Decision first**
   Verdict and one headline number; evidence on tap.
4. **Map that explains**
   Five tick-able layers, told apart by line style, not colour alone.
5. **Charts, not paragraphs**
   Area bars with ±0.2 dec band, lineage, tax, audit chain.
6. **Bangla / English, role-scoped**
   Digits follow language; fields follow the officer's role.

Fig 4. Annotated wireframes redrawn from the prototype (synthetic data): A, auto-verification; B, verdict panel for dag 402.

7 Five cross-agency consistency checks

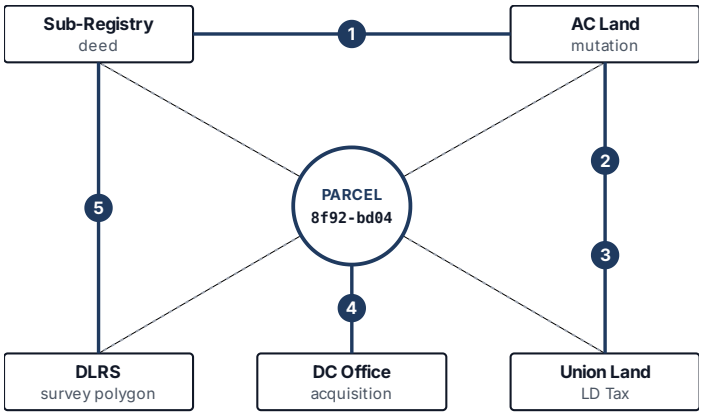


Fig 5. Each check compares two agencies' records of the same parcel.

#	Check	Agencies compared
1	Deed has a mutation (30 days)	Sub-Registry ↔ AC Land
2	Holder and payer names match (≥ 80%)	Union Land ↔ AC Land
3	LD Tax arrears within 1 year	Union Land (LD Tax)
4	No Khas / canal encroachment	Khas register ↔ deed
5	Deed area fits the polygon (±0.2 dec)	Sub-Registry ↔ DLRS

Verdict: CLEAR = records agree · REVIEW = needs a human look · BLOCK = records conflict. Naksha identifies discrepancies to support the authorised officer; it does not determine legal ownership or adjudicate disputes. Thresholds are configurable.

Missing or unreadable data is reported as UNKNOWN and is never treated as a pass. Planned checks: land-class agreement and acquisition-notice status.

8 System architecture

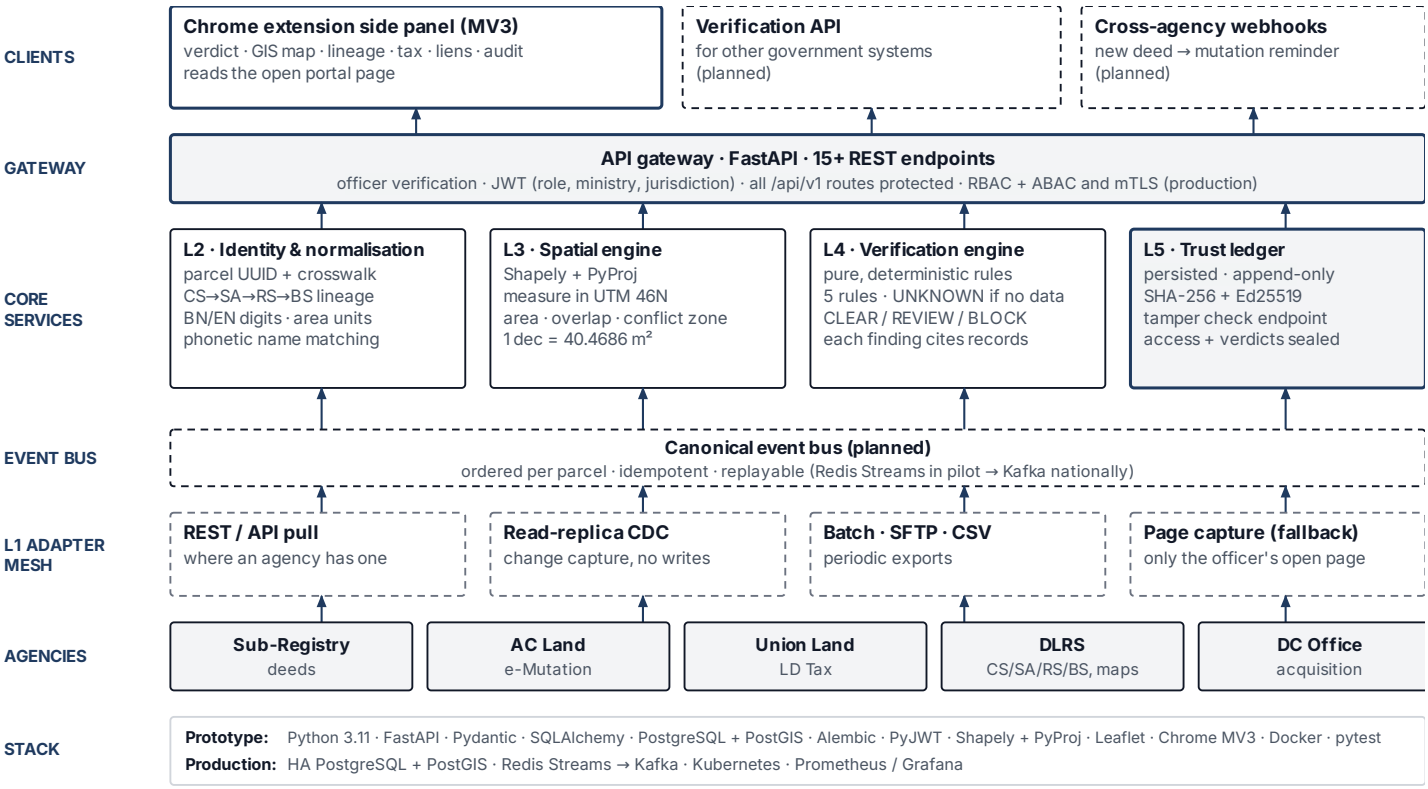


Fig 6. Five layers behind one gateway. Data flows up from agencies, which remain the source of record. Dashed boxes are planned.

9 Data interoperability: from mismatch to common data

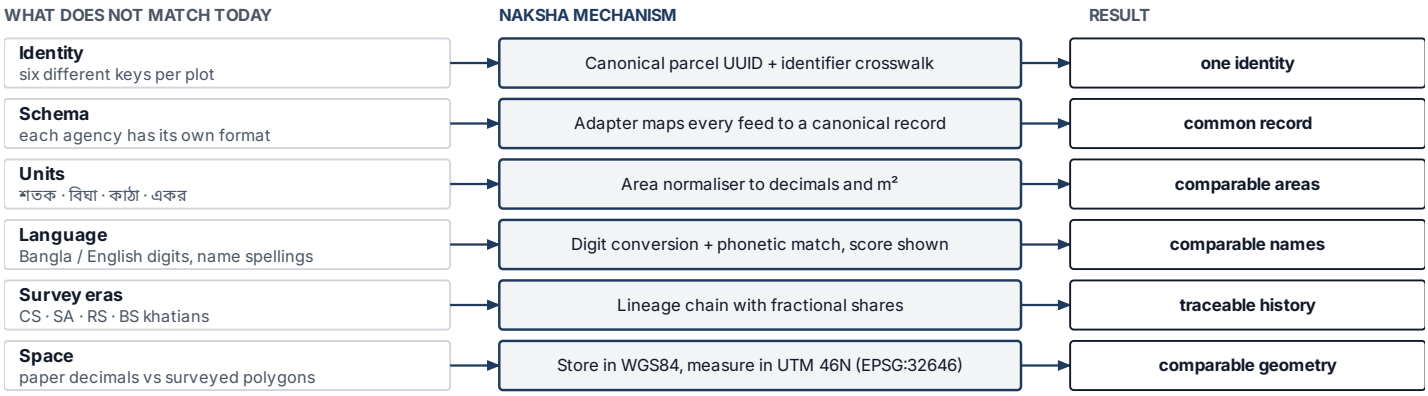


Fig 7. Six kinds of mismatch, each resolved by one mechanism, so records from different agencies become comparable.

10 Canonical data model

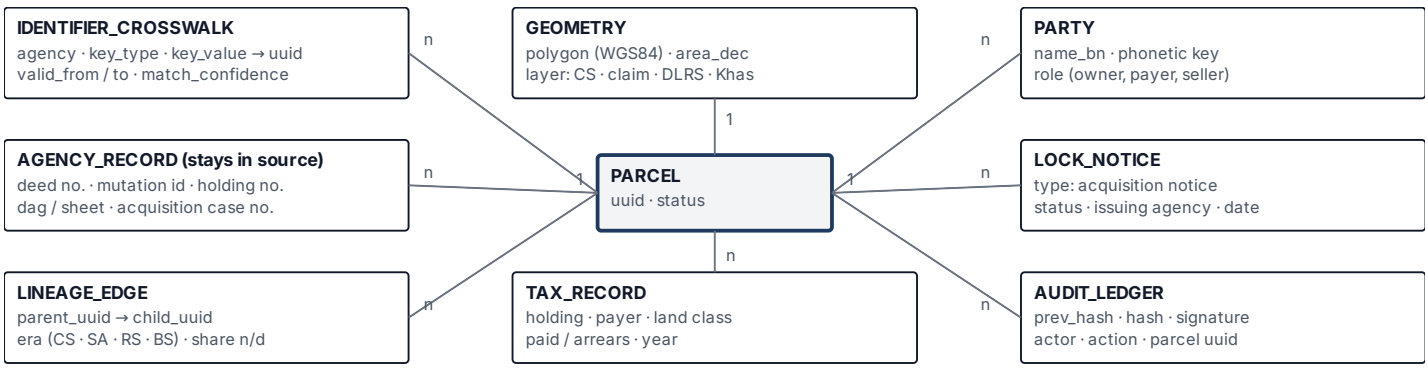


Fig 8. Everything hangs off one parcel UUID. Agency records are referenced, not copied.

Prototype tables: parcels, agency records, lineage, survey layers, Khas zones, tax, liens, officers, ledger (PostgreSQL + PostGIS).

11 How each agency connects

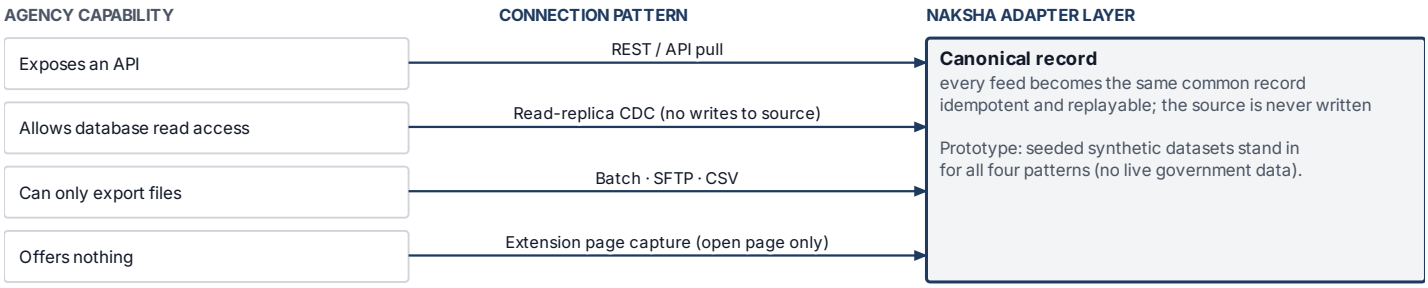
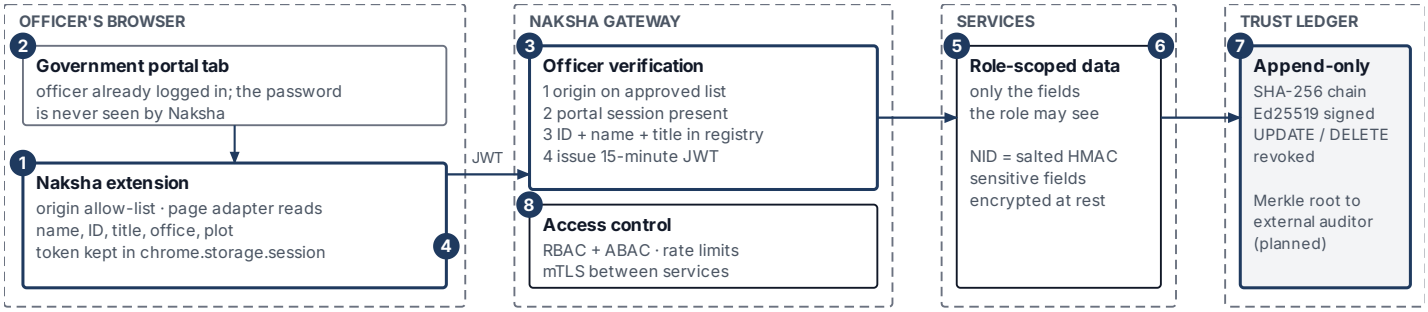


Fig 9. Adapter-based integration: the pattern follows what each agency can offer; adding an agency means writing one adapter. The prototype uses synthetic datasets; live integration is planned.

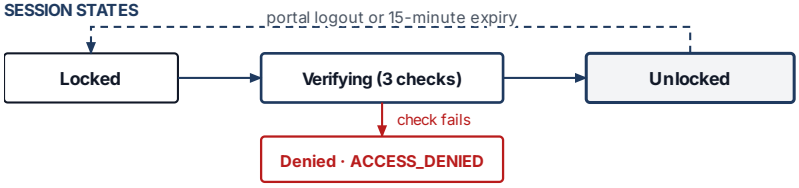
12 Security: verify the officer without another password



THREAT → CONTROL

1	**Wrong site:**	origin allow-list; nothing is injected elsewhere	5	**Over-sharing:**	role and jurisdiction scope every response
2	**Credential theft:**	no Naksha login; password never read or stored	6	**NID leak:**	no NID stored; IDs kept only as keyed HMAC hash
3	**Spoofed identity:**	active registry match: ID + name + title (prod: signed SSO)	7	**Log tampering:**	append-only table, hash chain, pinned-key signatures
4	**Stolen token:**	15-minute JWT, memory only, locks on portal logout	8	**Abuse:**	rate limit (built); mTLS (production)

SESSION STATES



Demo vs production trust

Demo: identity is read from a mock portal page and matched to a synthetic officer registry (simulated). Production: the portal SSO (OIDC) issues a signed token that Naksha validates; a page read alone can be spoofed.

Fig 10. Trust zones, numbered controls (1-8) and session states. The officer never enters a Naksha credential.

13 Prototype status and planned live integration

Component	Status	Note
Canonical models, normalisation (digits, area units, phonetics), lineage; PostGIS persistence (Docker), Alembic migrations	**Built**	Pydantic schemas; synthetic benchmark datasets
Spatial engine (UTM 46N), 5-rule engine, persistent trust ledger	**Built**	Shapely + PyProj + PostGIS; SHA-256 + Ed25519
Chrome MV3 side panel, officer verification gate, mock portals, 15+ REST endpoints, JWT auth with role + jurisdiction scope	**Built**	BN/EN, dark/light; 120+ automated tests
Agency data feeds, officer registry, SSO	**Simulated**	Seeded synthetic data; registry-based identity
Live agency adapters, event bus, SSO, mTLS, Merkle anchoring	**Planned**	Not built; no live government integration yet

14 Nationwide path and governance

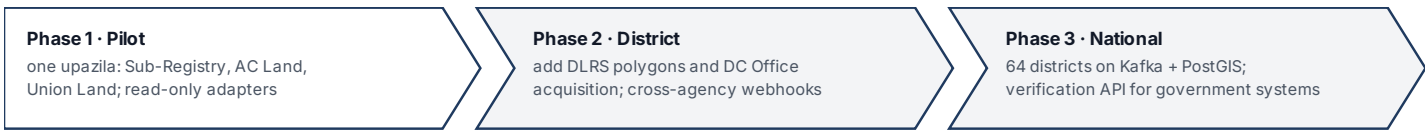


Fig 11. Rollout: work grows per agency (one adapter), not per district.

Proposed operating model: hosted with a government digital body (e.g. a2i or the Ministry of Land); each agency owns its records under a data-sharing agreement; live integration needs separate agreements.

15 Team and compliance

Team Leader	Team Member	Faculty Mentor	Industry Mentor
Mahadi Rahman Jihad 251-15-008 · CSE	Tartib Kaiser Munna 251-15-848 · CSE	Abdullah Al-Amin Lecturer, CSE	Farvez Rakib CSE-TECH

Synthetic data only; no real names, NIDs or land records; no scraping or testing of live government systems; AI tools used are listed in the README.